

Definite integrals – 4

Integrals with a parameter and passage to the limit

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Theoretical minimum.

Theorem 1 (continuity and the Leibniz rule). If $f(x, t)$ is continuous on $[a, b] \times [c, d]$, then $I(t) = \int_a^b f(x, t) dx$ is continuous. If in addition $\partial f / \partial t$ is continuous, then I is differentiable and $I'(t) = \int_a^b \partial_t f dx$; with variable limits, $\frac{d}{dt} \int_{\alpha(t)}^{\beta(t)} f(x, t) dx = \int_{\alpha}^{\beta} \partial_t f dx + \beta' f(\beta, t) - \alpha' f(\alpha, t)$.

Theorem 2 (uniform convergence). If integrable f_n converge to f uniformly on $[a, b]$, then f is integrable and $\int f_n \rightarrow \int f$. A uniformly convergent series may be integrated term by term.

Theorem 3 (Arzelà). If integrable f_n are uniformly bounded on $[a, b]$ and converge pointwise to an integrable f , then $\int f_n \rightarrow \int f$. Both hypotheses matter: without a uniform bound the conclusion fails, and without integrability of the limit the statement does not even make sense.

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1. Prove Theorem 1, including the version with variable limits, and prove Theorem 2 together with the statement about series.
 2. Prove Arzelà's theorem by an elementary argument, without measure theory, and show that the uniform bound cannot be dropped: construct integrable $f_n \rightarrow 0$ pointwise with $\int_0^1 f_n = 1$ for every n .
 3. (IMC) Let (f_n) be continuous on $[0, 1]$ with $\int_0^1 f_m f_n = 1$ for $m = n$ and 0 otherwise, and suppose $\sup\{|f_n(x)|\} < +\infty$. Show that no subsequence of (f_n) converges at every point of $[0, 1]$.
 4. (Proposed problems) Construct continuous $f_n: [0, 1] \rightarrow [0, 1]$ such that $\int_0^1 f_n \rightarrow 0$, the sequence $(f_n(x))$ converges for no $x \in (0, 1)$, and each f_n takes every positive value at most twice.

5. Prove that $I(t) = \int_0^1 |f(x) - t| dx$ is convex for every integrable f , that it is differentiable exactly where the distribution of f has no atom, and that its minimum is attained at a median of f .